Midterm Practice ____

- **Use the questions in this question bank to practice for the midterm test.**
- **Syllabus:** everything covered up to and including the week before the midterm test.

Reminders:

- The midterm counts for **20%** of your final grade.
- Both the midterm and the final test are **closed book.**
  - However, you are allowed to bring 1 A4 size sheet.
  - You are also allowed to bring your calculator.

Supervised Learning Questions:

1. What is the main goal of supervised learning?

A. To find patterns in data without labeled responses

B. To create labeled data from unlabeled data

C. To learn a mapping function from input features to output labels

D. To only perform clustering on data

2. In supervised learning, which of the following tasks is an example of regression?

A. Predicting the type of fruit based on color

B. Classifying emails as spam or not spam

C. Predicting the price of a house based on its features

D. Grouping customers into segments

3. Which of the following loss functions is commonly used in regression tasks?

A. Binary Cross-Entropy

B. Categorical Cross-Entropy

C. Mean Squared Error (MSE)

D. F1 Score

4. Which loss function is typically used for binary classification?

A. Mean Squared Error (MSE)

B. Mean Absolute Error (MAE)

C. Binary Cross-Entropy (Log Loss)

D. Accuracy

5. What does Mean Absolute Error (MAE) measure?

A. The total squared difference between predicted and actual values

B. The average of absolute differences between predicted and actual values

C. The percentage of correct predictions

D. The likelihood of a binary outcome

6. In the context of supervised learning, which of the following is NOT a typical step?

A. Defining the model structure

B. Computing model weights using an optimization algorithm

C. Labeling the dataset manually

D. Evaluating the model with a test set

7. Given a loss function $L(y, \hat{y})$, which of the following operations is central to the training phase in supervised learning?

A. Finding the predictor $h^* \in \mathcal{H}$ that maximizes L over D_{train}

B. Finding the predictor $h^* \in \mathcal{H}$ that minimizes L over D_{test}

C. Finding the predictor $h^* \in \mathcal{H}$ that maximizes L over D_{test}

D. Finding the predictor $h^* \in \mathcal{H}$ that minimizes L over D_{train}

8. In the context of supervised learning, classification aims to:

A. Predict continuous numerical values

B. Group similar items together without labels

C. Predict discrete labels based on input features

D. Create new features from unlabeled data

9. Which of the following describes the purpose of using a test set?

A. To train the model using labeled data

B. To evaluate the model's performance on unseen data

C. To optimize the model's parameters

D. To visualize the data trends

10. Which supervised learning method is suitable for predicting if an email is spam or not spam?

A. Regression

B. Clustering

C. Classification

D. Dimensionality Reduction

Supervised Learning Answers:

1. C

2. C

3. C

4. C

5. B

6. C

7. D

8. C

9. B

10. C

Linear Regression Basics:

1. What is the primary objective of linear regression?

A. To classify data points into groups

B. To establish a linear relationship between input and output variables

C. To reduce dimensionality of data

D. To perform clustering on data

Answer: B

2. In linear regression, what do we call the variable that we are trying to predict?

A. Feature

B. Independent variable

C. Target variable

D. Covariate

Answer: C

3. Which of the following is true for simple linear regression, i.e., 1D features and 1D labels?

A. There are multiple independent variables

B. The output variable must be binary

C. There is only one independent variable

D. It can only be used for classification tasks

Answer: C

4. What is the purpose of the intercept in a linear regression model?

A. It represents the slope of the line

B. It represents the point where the line crosses the y-axis

C. It defines the residual sum of squares

D. It is not used in linear regression

Answer: B

5. Which cost function is most commonly used in linear regression?

A. Cross-Entropy Loss

B. Mean Squared Error (MSE)

C. Hinge Loss

D. Log Loss

Answer: B

6. In linear regression, the slope represents:

A. The change in the dependent variable for a unit change in the independent variable

B. The maximum value of the dependent variable

C. The minimum value of the independent variable

D. The cost of prediction errors

Answer: A

7. What is residual error in linear regression?

A. The predicted value minus the actual value

B. The actual value minus the predicted value

C. The square of the predicted value

D. The mean of all predictions

Answer: B

8. Which of the following represents root mean squared error?

A. $L(y, \hat{y}) = \frac{1}{N_{train}} \sqrt{\sum_{i \in D_{train}} y_i - \hat{y}_i}$

B. $L(y, \hat{y}) = \frac{1}{N_{train}} \sqrt{\sum_{i \in D_{train}} |y_i - \hat{y}_i|}$

C. $L(y, \hat{y}) = \sqrt{\frac{1}{N_{train}} \sum_{i \in D_{train}} (|y_i - \hat{y}_i|)^2}$

D. $L(y, \hat{y}) = \frac{1}{N_{train}} \sqrt{\sum_{i \in D_{train}} (y_i - \hat{y}_i)^2}$

Answer: C

Linear Regression Using Scikit-Learn:

1. Which library in Python is primarily used for implementing linear regression models?

A. NumPy

B. Scikit-Learn

C. Matplotlib

D. Pandas

Answer: B

2. In Scikit-Learn, which class is used to create a linear regression model?

A. `LinearModel()`

B. `Linear_Regression()`

C. `LinearRegression()`

D. `RegressionModel()`

Answer: C

3. What does the `fit()` method in Scikit-Learn's `LinearRegression` class do?

A. It initializes the model

B. It trains the model using the given dataset

C. It saves the model to a file

D. It evaluates the model on test data

Answer: B

4. Which function in Scikit-Learn is used to split data into training and test sets?

A. `train_test_split()`

B. `split_data_train_test()`

C. `data_split()`

D. `split_train_test()`

Answer: A

5. In Scikit-Learn, what is the default evaluation metric for linear regression?

A. Accuracy

B. Precision

C. Mean Squared Error

D. Recall

Answer: C

6. Which method in the `LinearRegression` class is used to make predictions on new data?

A. `predict()`

B. `fit()`

C. `evaluate()`

D. `split()`

Answer: A

7. How does one compute $L(y, \hat{y}) = \frac{1}{N_{test}} \sqrt{\sum_{i \in D_{test}} |y_i - \hat{y_i}|}$ in Scikit-Learn's `LinearRegression`?

A. `sklearn.error_metrics.mean_squared_error()`

B. `sklearn.preprocessing.mean_absolute_error()`

C. `sklearn.metrics.mean_absolute_error()`

D. `sklearn.metrics.mean_absolute_error()`

Answer: C

8. Which parameter in `` `train_test_split()` `` determines the size of the test set?

A. `test_size`

B. `size`

C. `test_ratio`

D. `split_ratio`

Answer: A

9. In Scikit-Learn, which attribute of `LinearRegression` holds the coefficients of the model?

A. `coef_`

B. `coefficients_`

C. `weights_`

D. `beta_`

Answer: A

10. Which attribute in Scikit-Learn's LinearRegression class represents the intercept?

A. `bias_`

B. `intercept_`

C. `alpha_`

D. `constant_`

Answer: B

11. What is the purpose of using `` `train_test_split()` `` in model training?

A. To shuffle the data

B. To split the data into training and test datasets

C. To evaluate the model on the training set

D. To load the data into memory

Answer: B

12. Which library does Scikit-Learn often use to handle numerical operations?

A. Matplotlib

B. NumPy

C. Pandas

D. math

Answer: B

13. What is the typical purpose of the `random_state` parameter in `train_test_split()`?

A. To make the data shuffle random

B. To ensure reproducibility of results

C. To set the size of the training set

D. To normalize the data

Answer: B

14. Which of the following methods is used to evaluate the accuracy of a linear regression model in Scikit-Learn?

A. `accuracy_score()`

B. `mean_squared_error()`

C. `confusion_matrix()`

D. `recall_score()`

Answer: B